

EXERCISE 4 · AFTER DECK 2

Guardrails, Strands, and End-to-End Agent Design

Synthesis — design a working agent and reason about what can go wrong.

6 questions · ~15–20 minutes · Worksheet only · answer key in separate PDF

WHAT TO DO

Six questions. About 20 minutes. The final design scenario is open-ended; answer it the way you'd answer in a real architecture review. There's no single right answer — the reasoning is the point.

Q1 OF 6 · CONCEPT

True or false: when you use the Strands Agents SDK, it bypasses the Converse API and uses a different, more efficient protocol to talk to Bedrock.

- A. True — Strands uses a custom binary protocol for lower latency.
- B. False — Strands calls the same Converse API underneath; it just orchestrates the loop for you.
- C. True — Strands talks directly to the model server, skipping Bedrock entirely.
- D. False — Strands only works with Anthropic models, not Bedrock.

Your answer: _____

Q2 OF 6 · BUG SPOT

A team configured a Guardrail with strict content filters. Their app started blocking legitimate questions. They want to relax the filters for testing. What is the **wrong** way to do this?

- A. Create a new version of the guardrail with looser thresholds and point dev to the new version.
- B. Edit the existing guardrail's thresholds in place and keep using the same version number.
- C. Configure two guardrails — one strict for prod, one loose for dev — and reference different IDs per environment.
- D. Leave the guardrail unchanged and disable it in dev by not passing the guardrail config.

Your answer: _____

Q3 OF 6 · BUG SPOT

Spot the issue in this Strands agent setup. The agent runs but never seems to use the knowledge base for any query.

```
from strands import Agent
from strands.models import BedrockModel
from strands_tools import retrieve

# KB ID not set anywhere

model = BedrockModel(
    model_id="us.amazon.nova-lite-v1:0",
    region_name="us-east-1",
)

agent = Agent(
    model=model,
    tools=[retrieve],
    system_prompt="You are a helpful assistant.",
)
```

Write your answer below:

Q4 OF 6 · CONCEPT

Looking at the before/after collapse table in Chapter 8, which of the following is **NOT** a thing the Strands Agent class handles for you?

- A. Appending the model's response to the conversation history
- B. Dispatching tool calls to the right Python function
- C. Calling Bedrock from a region your code doesn't have access to
- D. Applying the guardrail to every model call

Your answer: _____

Q5 OF 6 · CONCEPT

You see this pattern in production code: the same Guardrail ID is referenced from a plain Converse call, a Knowledge Base query, *and* a Strands agent's BedrockModel. What's the implication?

- A. Bad practice — guardrails should be unique per API surface.
- B. Good practice — one guardrail, configured once, enforced everywhere; safety stays consistent across the application.
- C. Will throw an error — guardrails can only attach to one call type at a time.
- D. Will silently fall back to no guardrail on whichever call ran second.

Your answer: _____

Q6 OF 6 · DESIGN SCENARIO

You're asked to build an agent for an HR team. It should answer policy questions (leave, expenses, benefits) from internal docs, and let employees check their own remaining vacation days via a backend API.

Sketch the agent in 3-5 lines: which tools, which guardrails, and one risk you'd flag to the security team.

Write your answer below:
